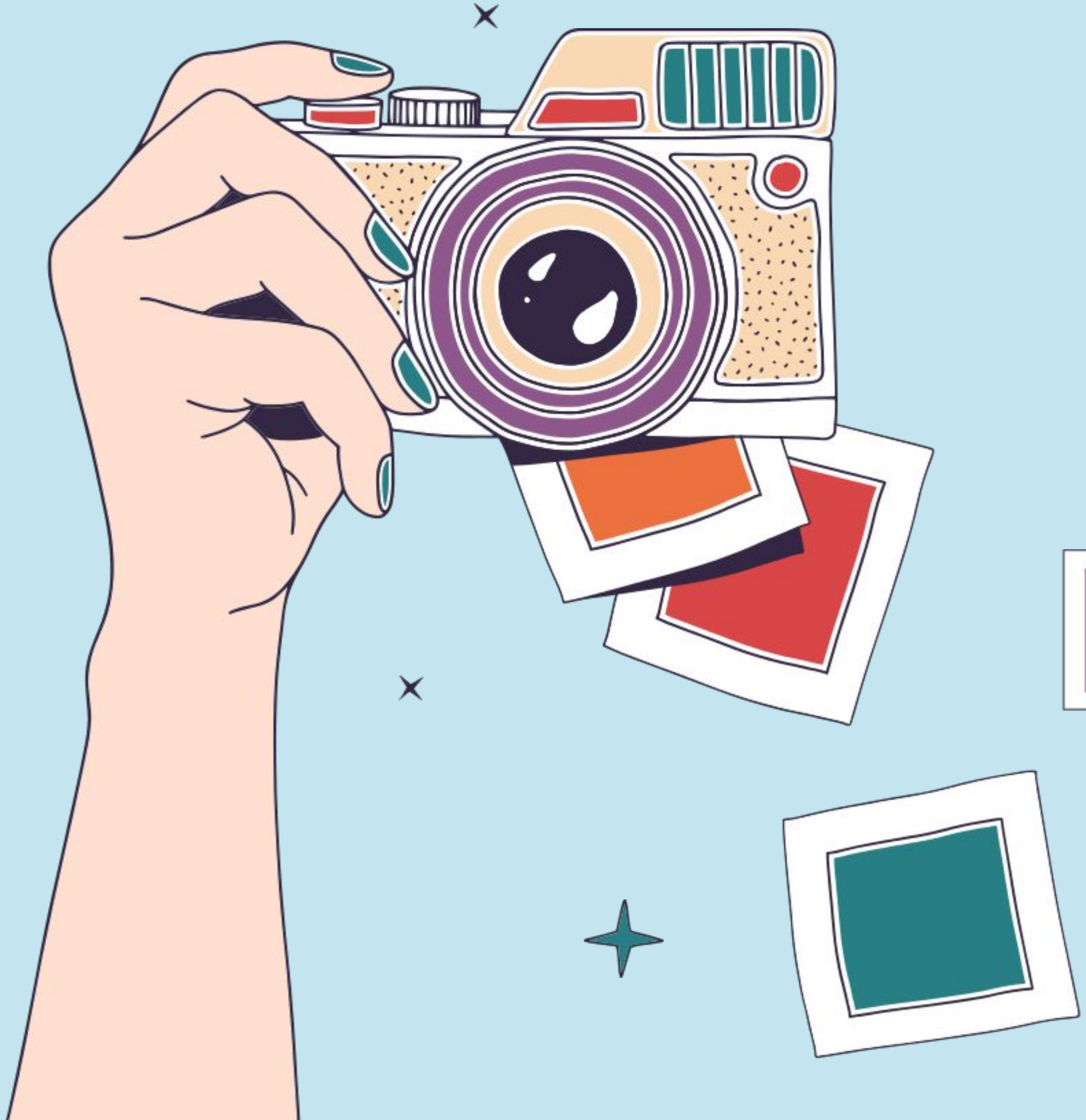


FRAME

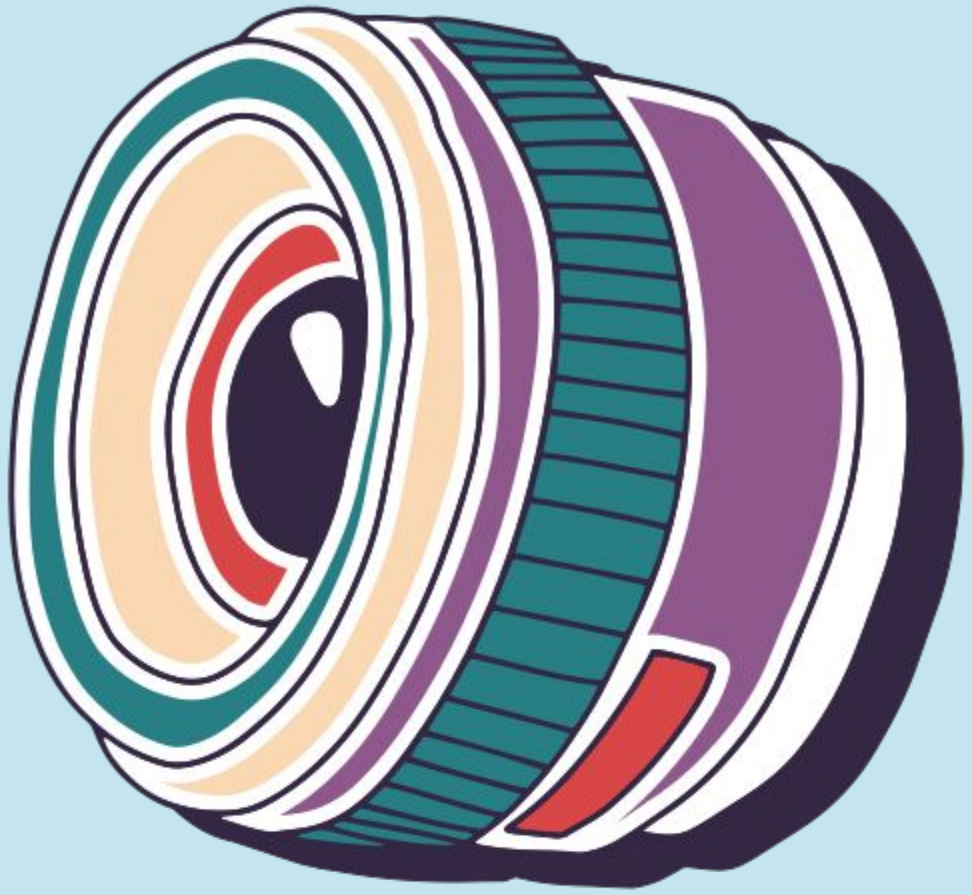
Memories shouldn’t be work!

table of content

table of content

01

intro



problem statement

- Photo sharing breaks down after the moment, turning memories into manual labor involving sorting, app-switching, and follow-ups instead of enjoyment.

- People lose confidence once photos are shared, worrying about who has access, how long they'll have it, and whether boundaries are respected over time.

- The burden falls on one "default sharer", who becomes a human bridge across iOS and Android, sacrificing time and mental energy to keep everyone included.

FRAME

SHARING MEMORIES MADE SIMPLE

Value Proposition Statement

Frame transforms scattered photos into shared, organized memories by automatically collecting, curating, and synchronizing albums across groups

Solution

Frame turns scattered event photos on iOS or Android into unified shared albums automatically. Frame combines intelligent event clustering, privacy-focused facial detection, and cross-platform synchronization to remove the burden of sorting and ensure memories reach the intended audience effortlessly.

our proposed solution is cross-platform intentional sharing

solution

Frame is a cross-platform photo sharing experience designed to keep memories effortless after the moment ends. It removes the burden of manual sorting, app-switching, and follow-ups by automatically creating shared albums and making audience visibility intentional and clear. Instead of turning one person into the "default sharer," Frame helps groups share photos seamlessly across iOS and Android without losing trust, time, or momentum.

The people behind FRAME

GAVIN CAO

loves animals, food & nature
lowkey a farmer :)

RACHIT GUPTA

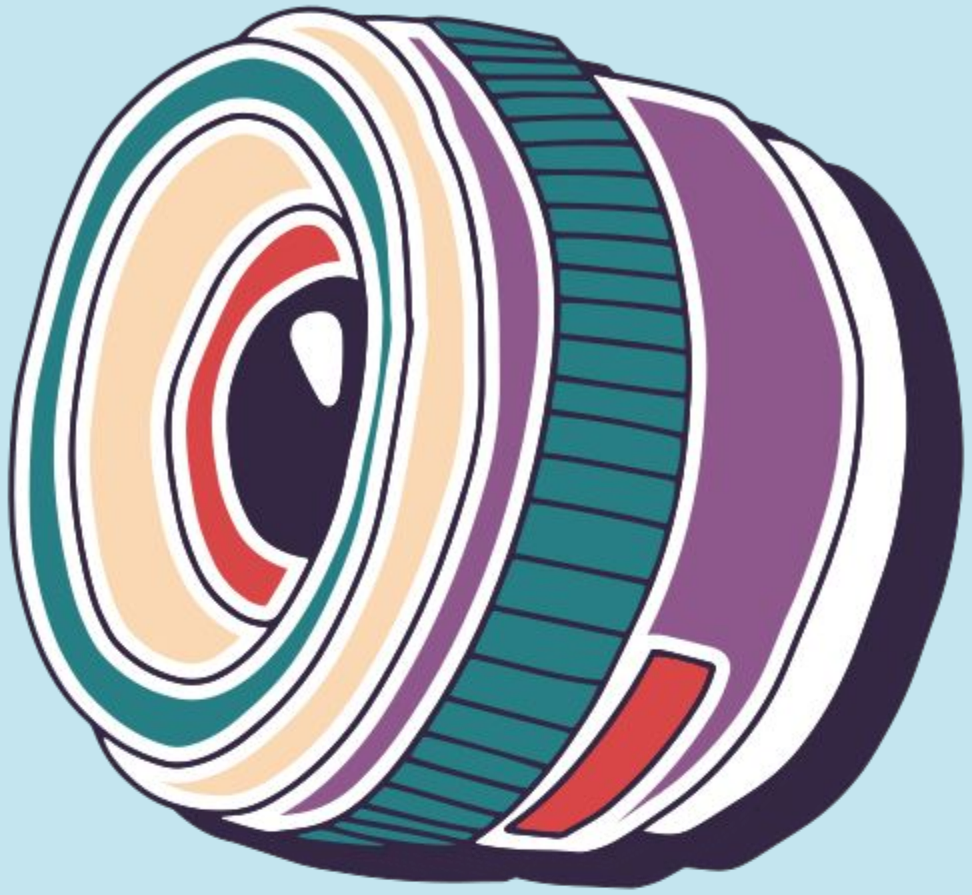
loves to drive, travel, and
click pictures!!

ZEEL PATEL

loves to sleep, eat
and repeat

02

solution exploration



solutions

Phone: Event Auto-Album Creation & Sharing
User finds a share-ready album automatically created from an event that can then be shared with intended audience

AR Glasses: Location-Based Memory Viewer
User revisits a place and sees floating albums tied to memories captured there.

Phone: Selective Audience Sharing
User sends different versions of the same album to friends, family, or private groups

Watch: Proximity Share Trigger
When friends nearby also attended the event, watch prompts quick sync

Desktop: Collaborative Album Editing
Friends co-edit a shared album remotely like Figma for memories

Phone: Gallery-Integrated Smart Sharing
User selects photos in their native gallery and sends them directly to people detected in the images

solutions

Phone: Live Collaborative Upload
During an event, users upload photos into a shared album in real time as memories happen

AR Glasses: Live Group Memory Space
During an event, users wearing AR glasses see a shared memory space where photos taken by everyone appear in real time.

AR Glasses: Immersive Memory Reliving
User browses shared memories in a spatial gallery around them

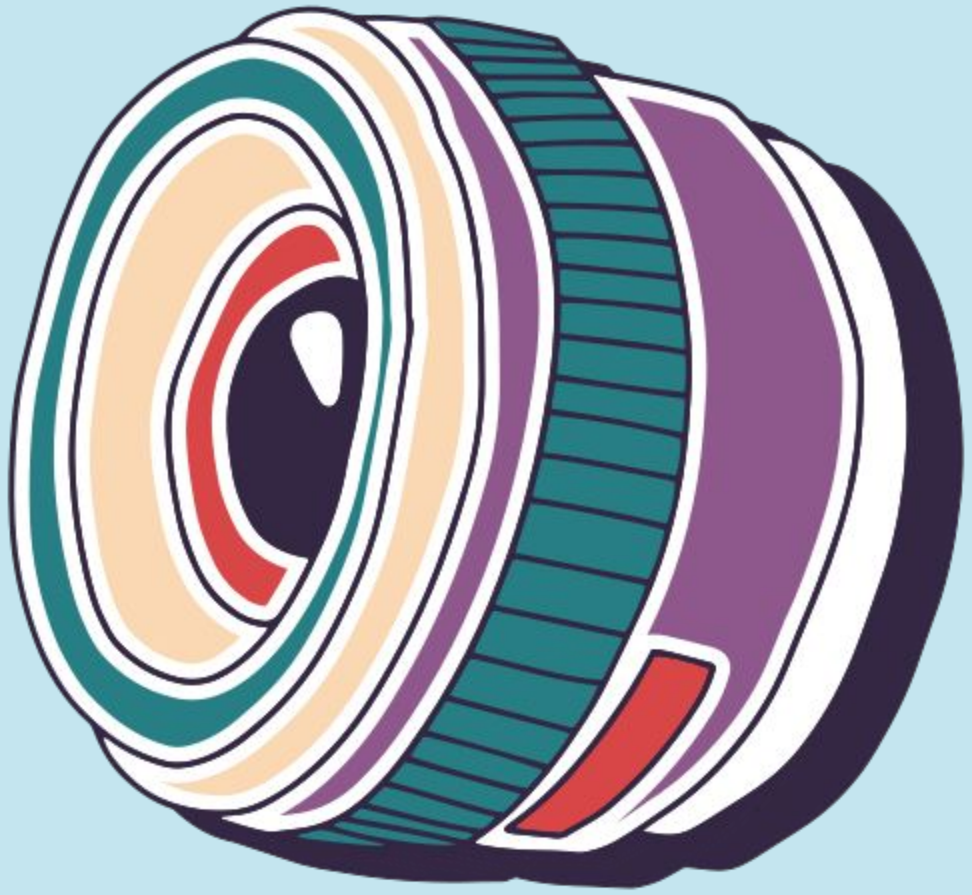
Desktop: Drag-and-Drop Memory Distribution Workspace
User uploads photos and drags them into audience zones like Friends, Family, or Private to distribute memories intentionally.

Phone: Temporary Memory Sharing
User shares party photos that automatically expire after a chosen time window.

Tablet: Group Album Curation Table
Friends sit together after a trip and collaboratively sort photos into a shared album which is better with a bigger screen

03

sketches

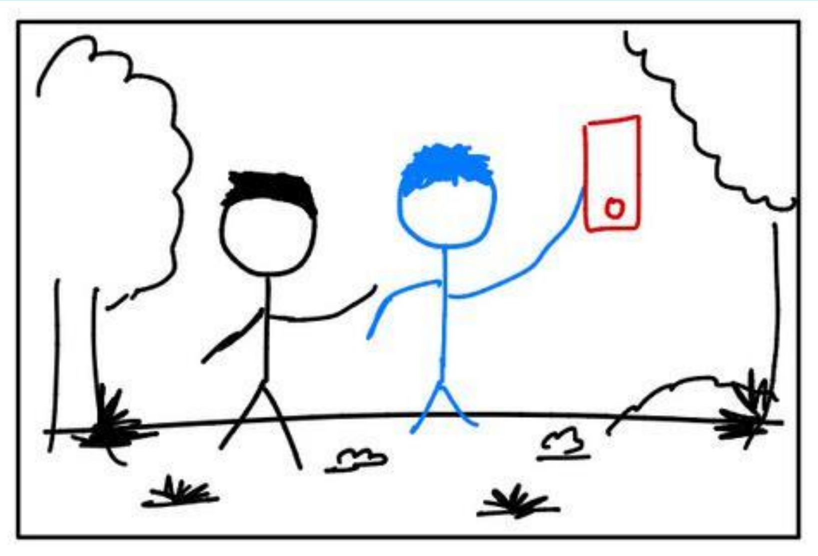


final concept sketches

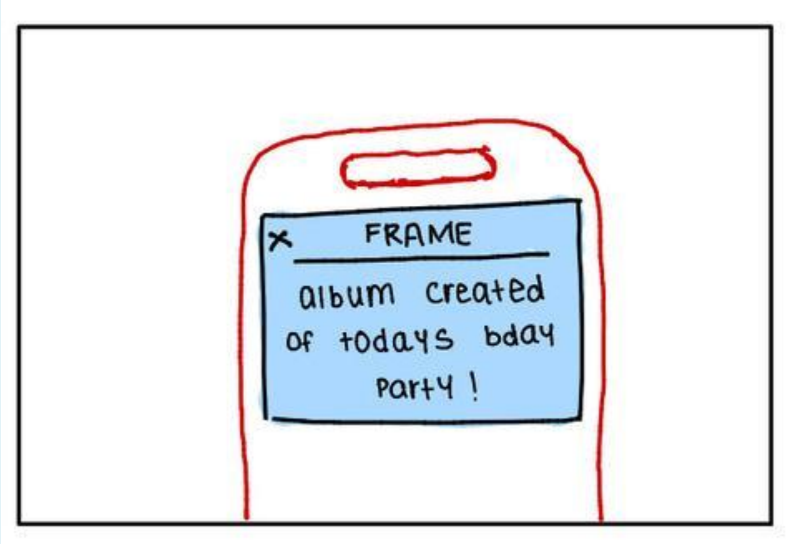
- Photo Phone: Event Auto-Album Creation & Sharing
- Desktop: Collaborative Album Editing
- AR Glasses: Location-Based Memory Viewer

Phone: Event Auto-Album Creation & Sharing

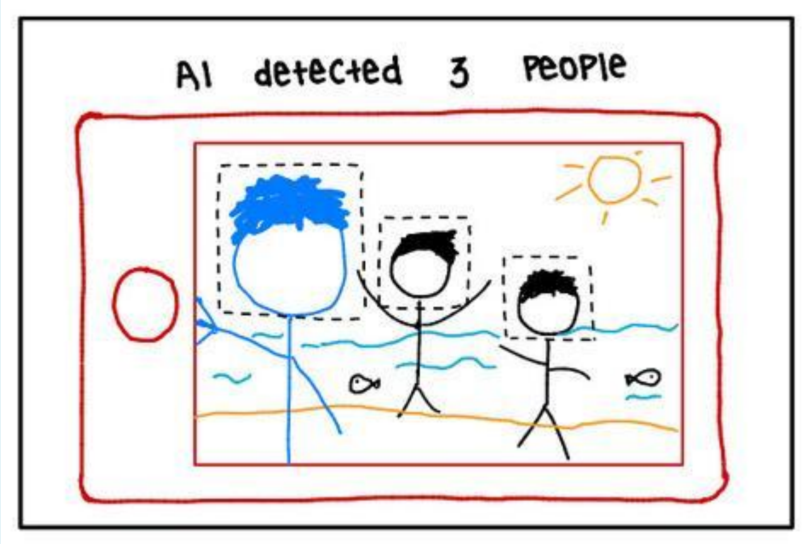
User finds a share-ready album automatically created from an event that can then be shared with intended audience



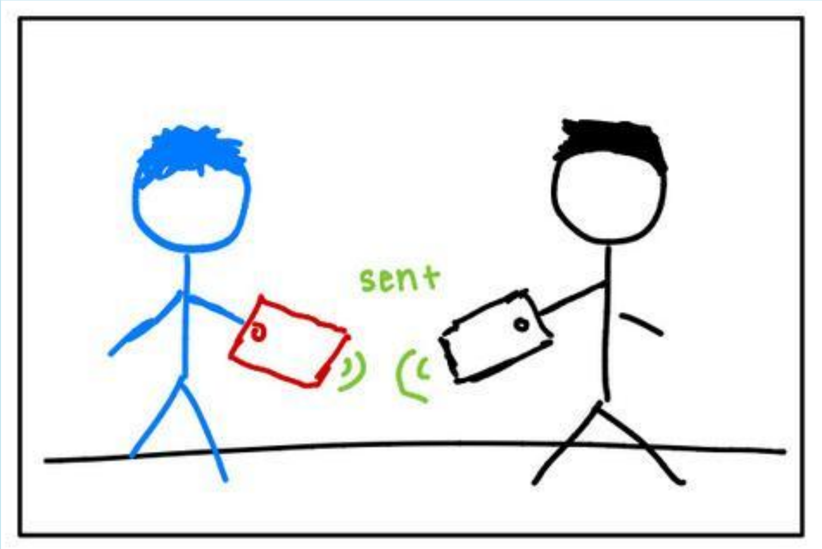
User allows gallery access so the app can gather event photos



User reviews an auto-created album based on shared memories



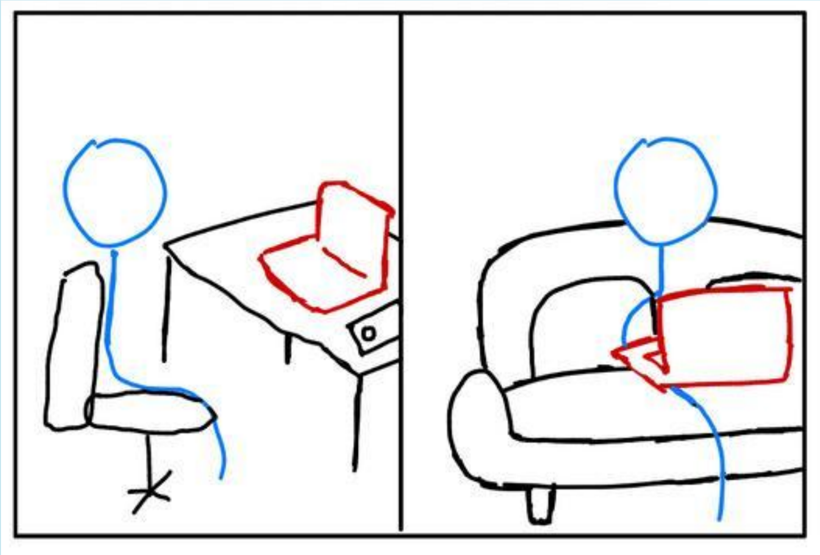
User sees suggested recipients based on people detected in the photos



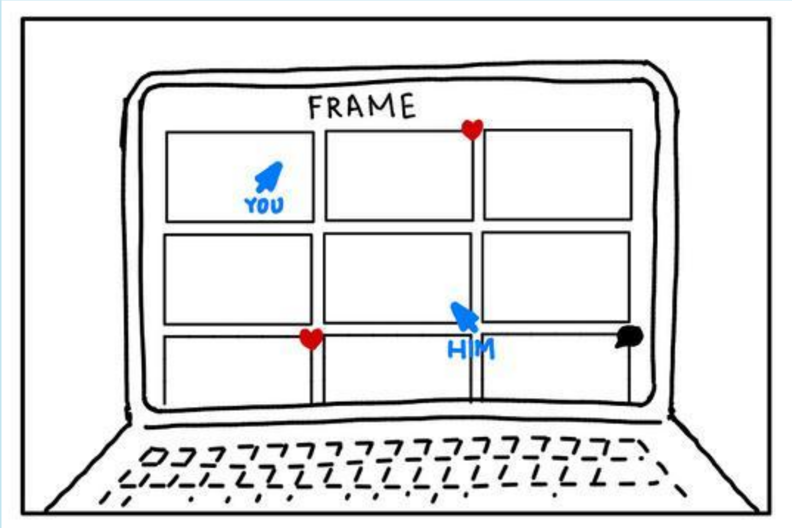
User can do cross-platform album delivery so everyone can access it

Desktop: Collaborative Album Editing

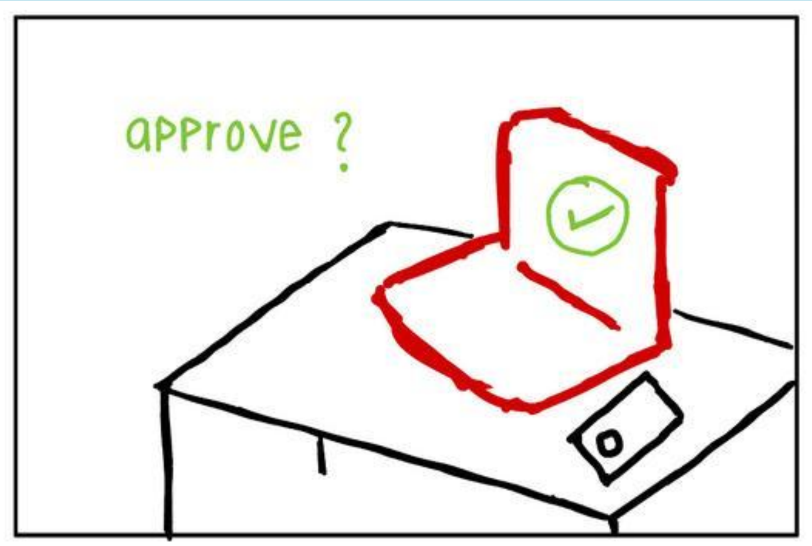
Friends co-edit a shared album remotely like Figma for memories



Friends access a shared album together from different locations



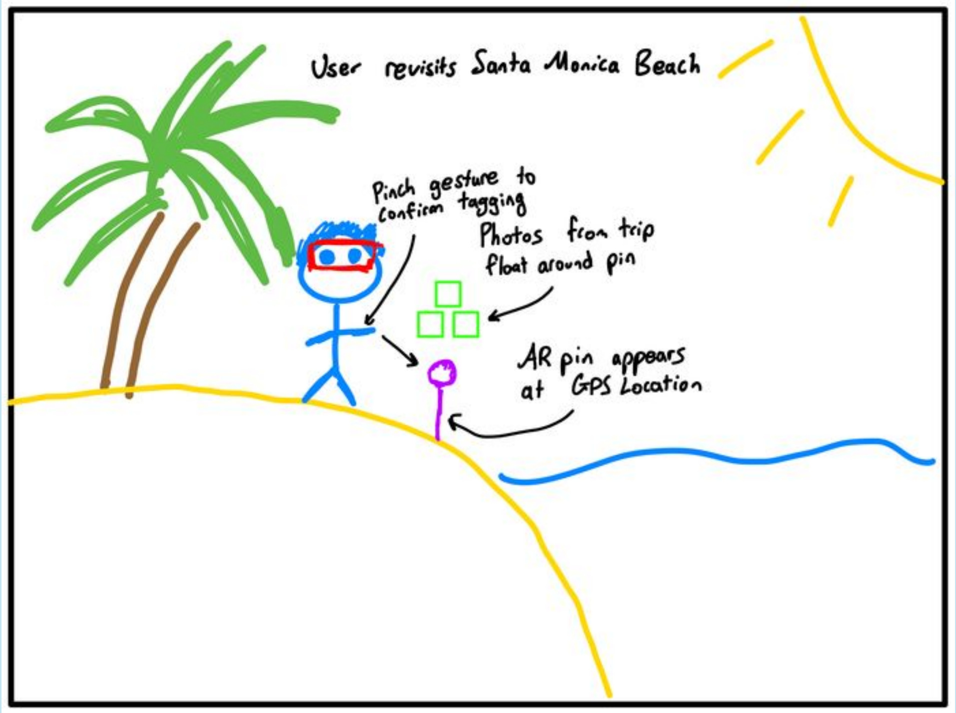
Friends organize and curate photos collaboratively in real time



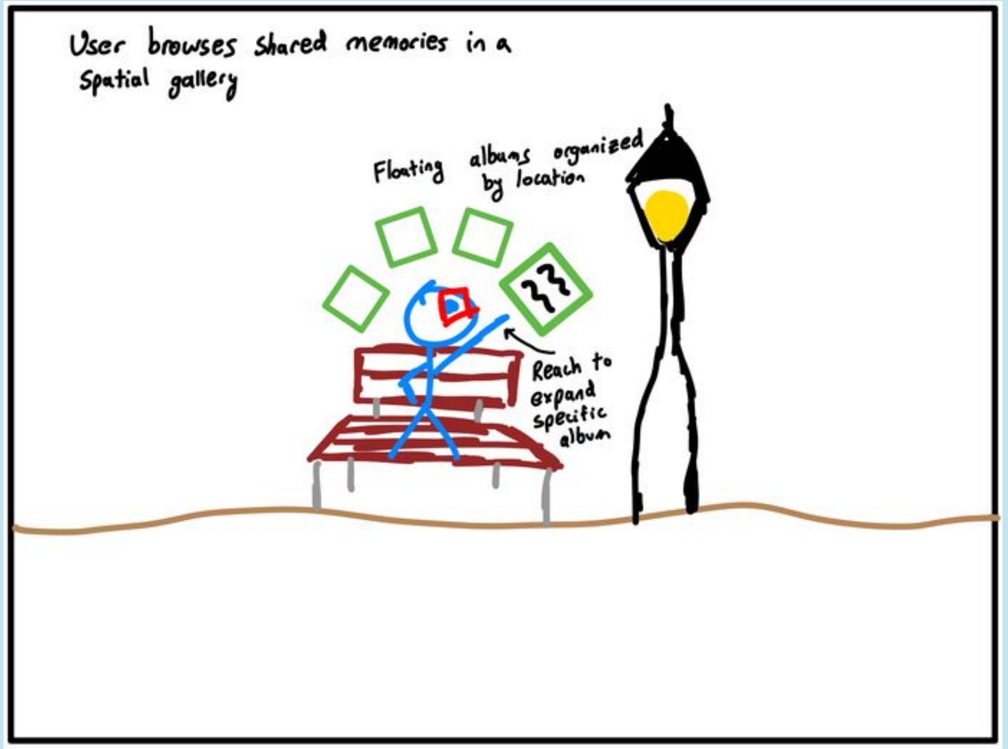
Friends review and finalize the album before sharing it

AR Glasses: Location-Based Memory Viewer

User revisits a place and sees floating albums tied to memories captured there.



Friends access a shared album together from different locations



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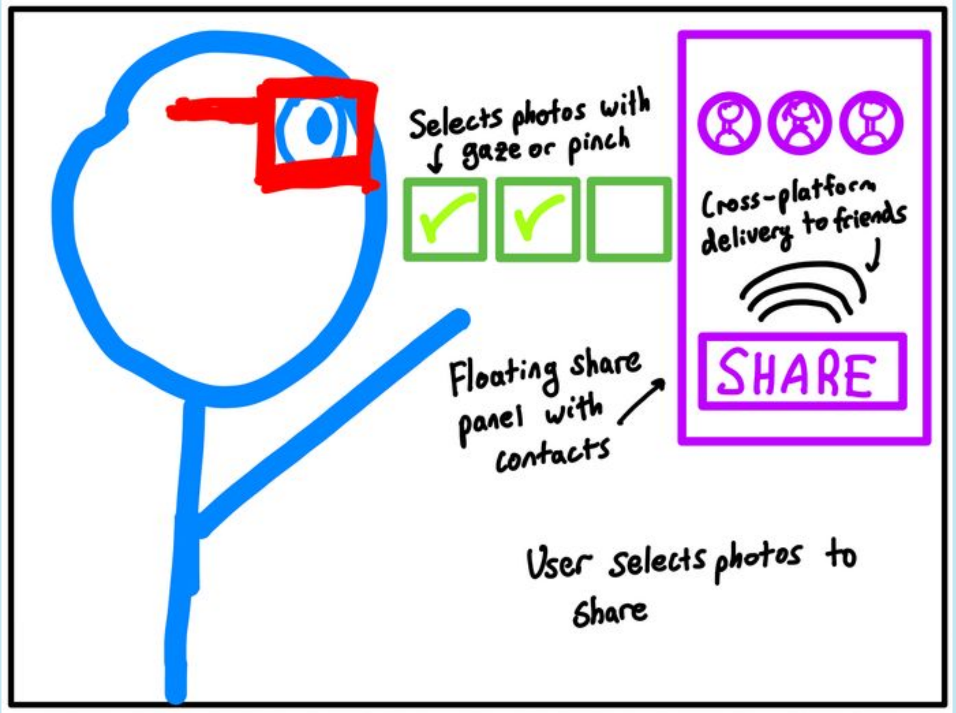


Friends review and finalize the album before sharing it

final key screens selections

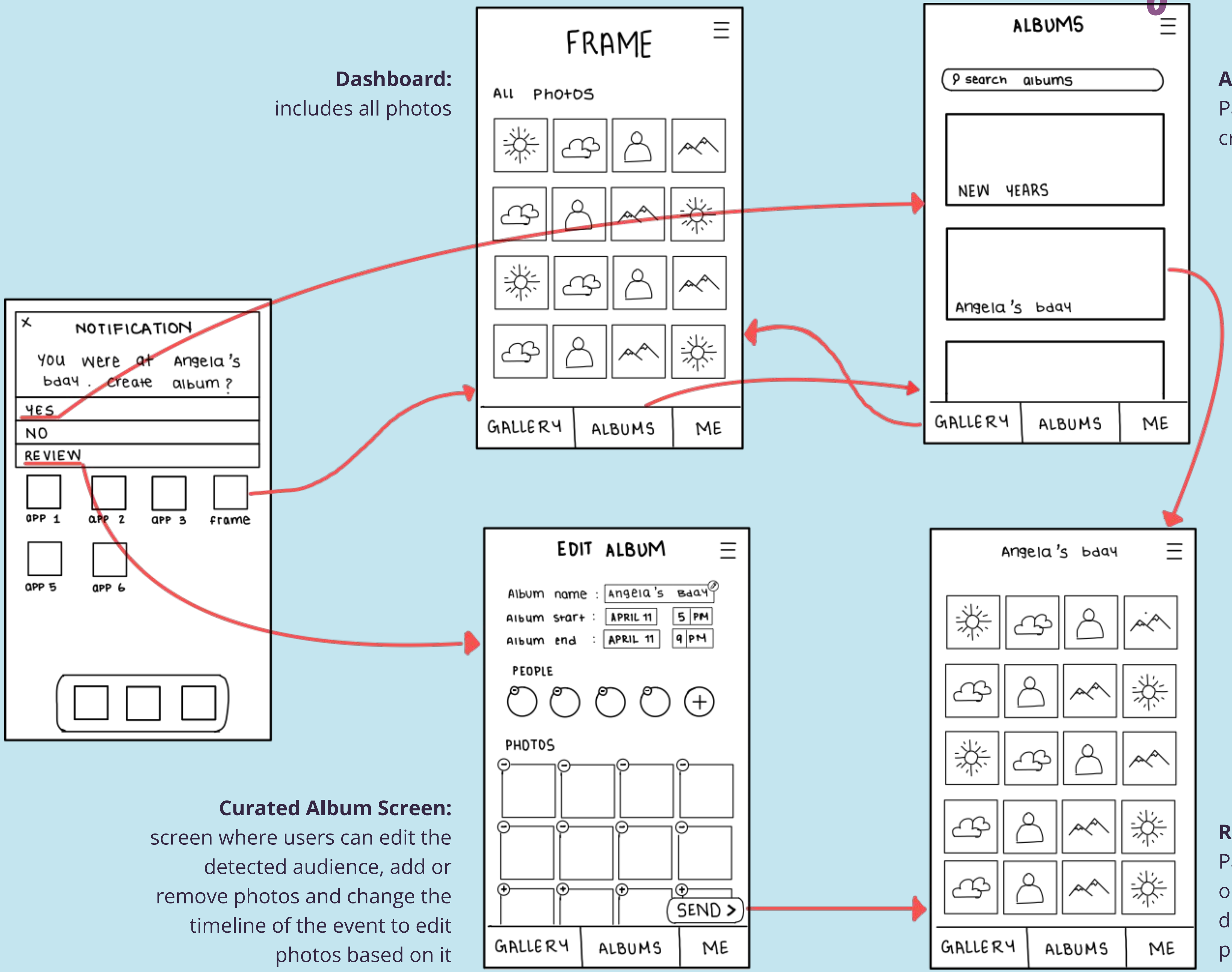
- Photo Phone: Event Auto-Album Creation & Sharing
- AR Glasses: Location-Based Memory Viewer

Phone: Event Auto-Album Creation & Sharing

Dashboard:
includes all photos

Albums Screen:
Page for displaying all created or recieved albums

Notification:
notified after automatic album creation, where the user has the option to review the album,send it or not create the album



Curated Album Screen:
screen where users can edit the detected audience, add or remove photos and change the timeline of the event to edit photos based on it

Received Albums Screen:
Page shows relevant photos or photos where the user is detected on top and other photos below it

Phone: Event Auto-Album Creation & Sharing

pros

- Lowest behavioral friction compared to all alternatives - Unlike AR glasses, watches, or desktop, this solution works within an already existing user habit, taking photos on their phone.

- No dependency on synchronous participation - Compared to live collaborative uploads or real-time memory spaces, this concept works even if participants take photos independently and at different times.

- Eliminates manual labor present in other phone and desktop concepts - Solutions like drag-and-drop distribution workspaces or collaborative editing require intentional effort, whereas auto-album creation shifts to the system.

- Greater scalability for large social events - Watch proximity sharing or tablet co-curation work best in small group settings, while automated clustering

cons

- Heavier reliance on AI inference - Unlike proximity triggers or manual sorting, this system depends on facial recognition and clustering accuracy to function well.

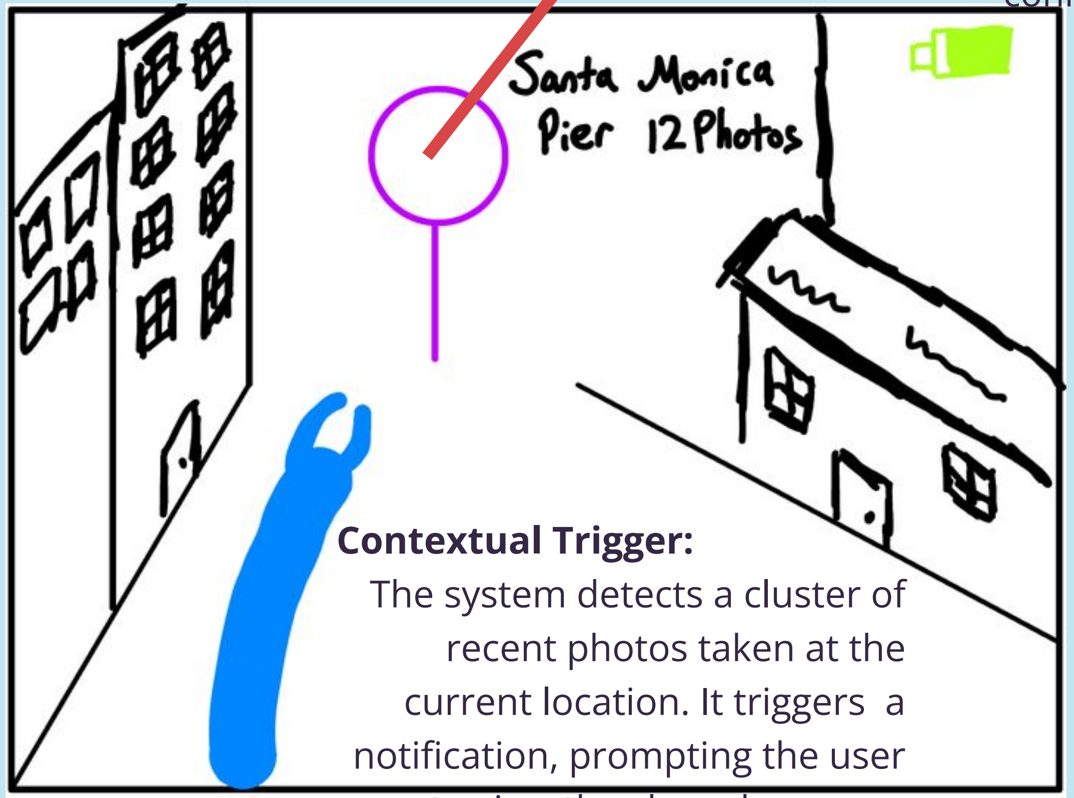
- Less intentional than manual curation tools - Desktop editing workspaces and drag-and-drop distribution allow precise storytelling, whereas automation may feel less personally crafted.

- Lower experiential novelty compared to AR solutions - Location-based viewing and immersive spatial galleries offer memorable, futuristic interactions that this phone solution does not.

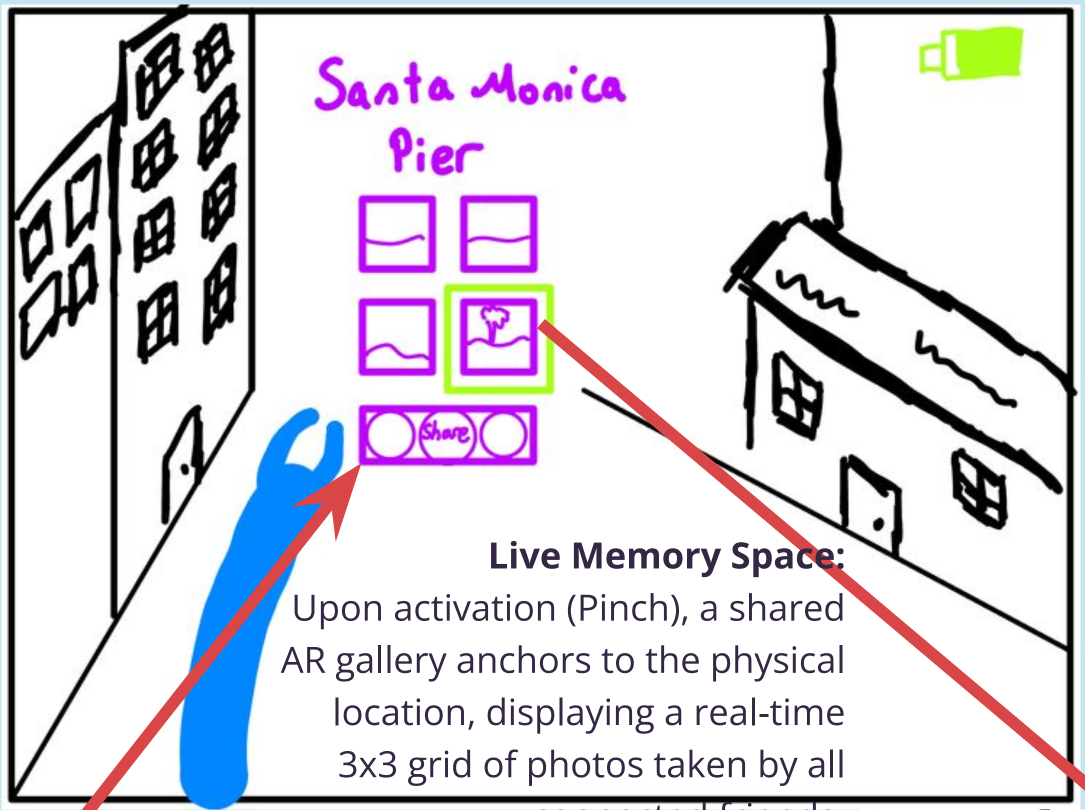
- Less playful or socially performative - Tablet curation tables or collaborative editing sessions create shared social activities, while automation is largely invisible.

AR Glasses: Live Group Memory Space

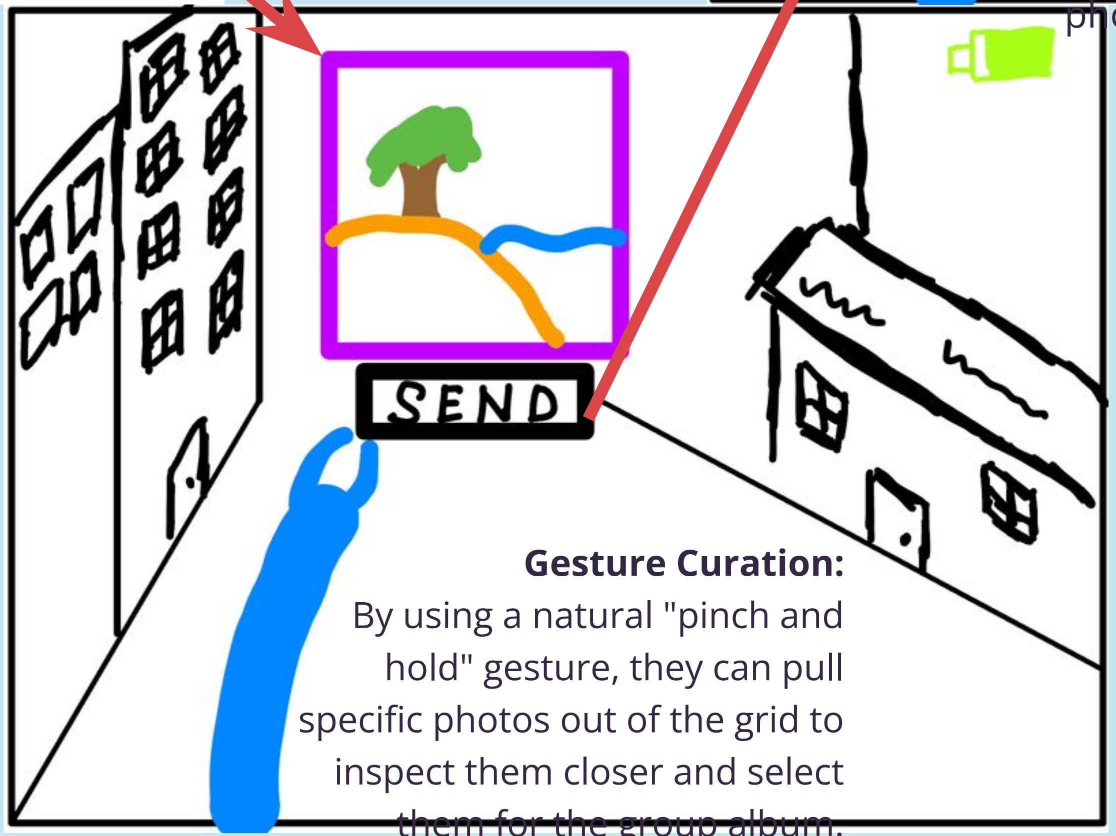
During an event, users wearing AR glasses see a shared memory space where photos taken by everyone appear in real time.



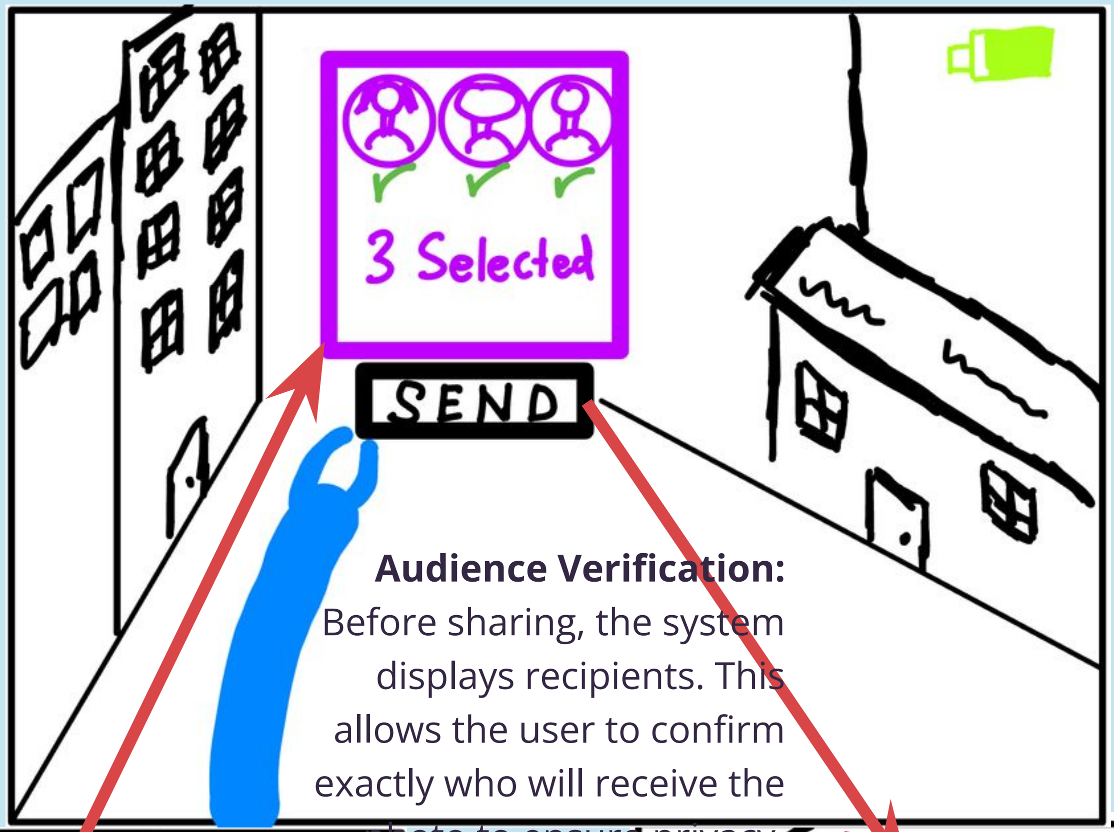
Contextual Trigger:
The system detects a cluster of recent photos taken at the current location. It triggers a notification, prompting the user to view the shared memory



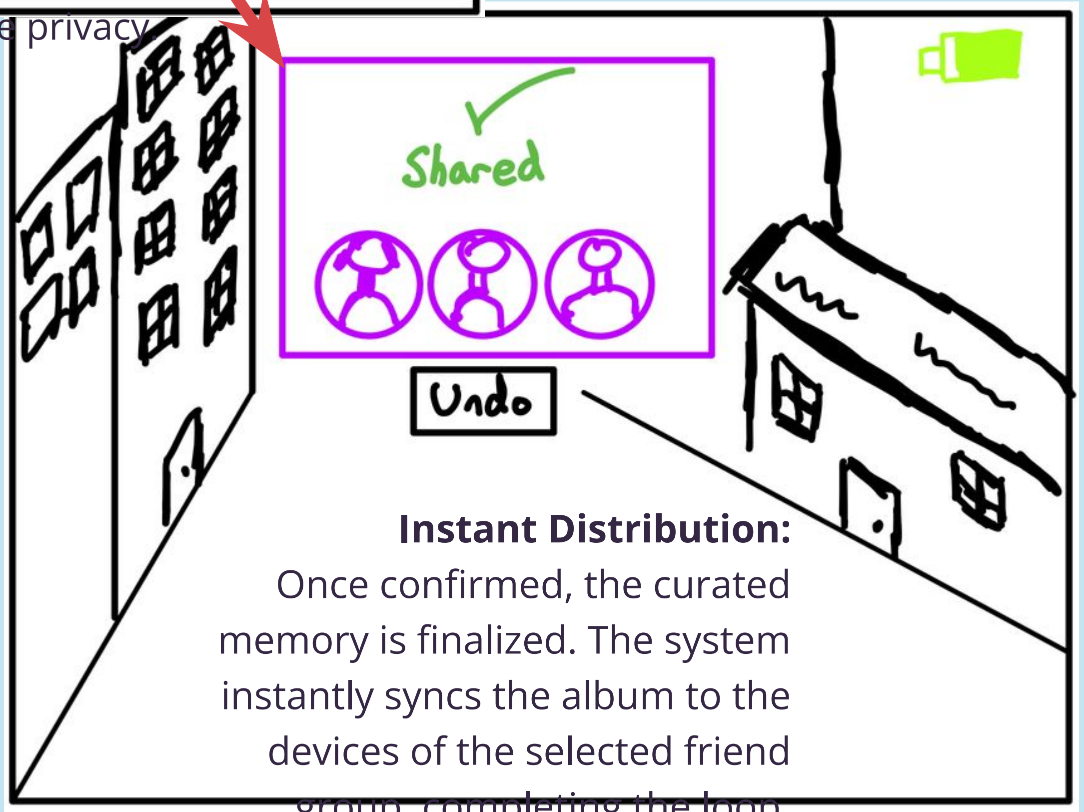
Live Memory Space:
Upon activation (Pinch), a shared AR gallery anchors to the physical location, displaying a real-time 3x3 grid of photos taken by all connected friends.



Gesture Curation:
By using a natural "pinch and hold" gesture, they can pull specific photos out of the grid to inspect them closer and select them for the group album.



Audience Verification:
Before sharing, the system displays recipients. This allows the user to confirm exactly who will receive the photo to ensure privacy.



Instant Distribution:
Once confirmed, the curated memory is finalized. The system instantly syncs the album to the devices of the selected friend group, completing the loop.

AR Glasses: Live Group Memory Space

pros

- Contextual Immersion: Viewing memories at the exact location they occurred creates a stronger emotional connection than scrolling through a disconnected gallery on a phone.
- Hands-Free Experience: Solves the "app switching" pain point by allowing users to interact with memories without breaking engagement with the real world (e.g., while hiking).
- Spatial Mental Model: Leveraging spatial memory reduces the cognitive load of searching through folders.
- Novelty & Differentiation: Fulfills the assignment goal of "novel realizations," distinguishing Frame from standard competitors like Google Photos or iCloud.

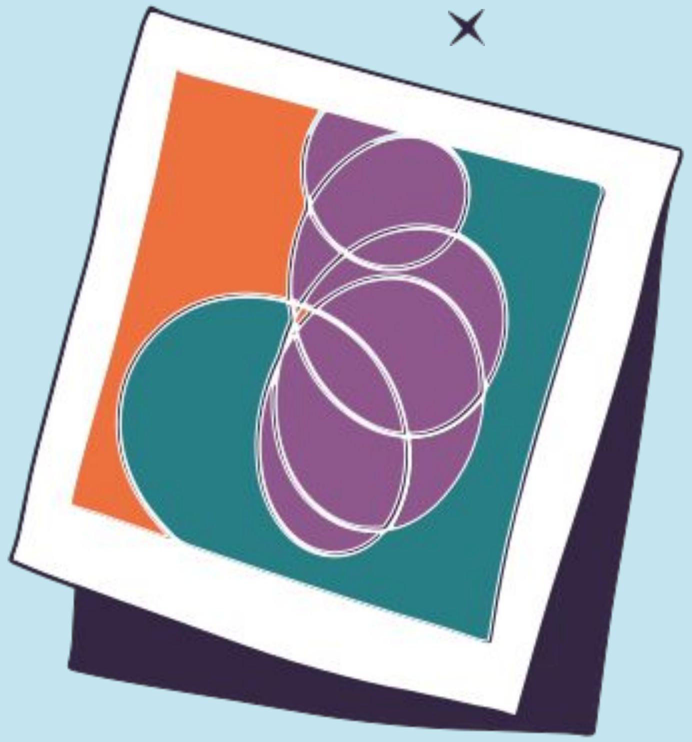
cons

- High Barrier to Entry: The target user base (students/young adults) generally does not own AR glasses, which severely limits adoption compared to a smartphone app.
- Location Dependency: The core value fails if users cannot physically revisit a location (e.g., a one-time vacation spot). A phone app works anywhere.
- Social Friction: Gesturing at invisible objects in public can cause social awkwardness and raise privacy concerns for bystanders who don't know if they are being recorded.
- Lack of Tactile Feedback: "Air tapping" is less precise than a phone screen, potentially making simple tasks (like selecting 1 specific photo) more frustrating and error-prone.

our chosen solution is Phone: Event Auto-Album Creation & Sharing

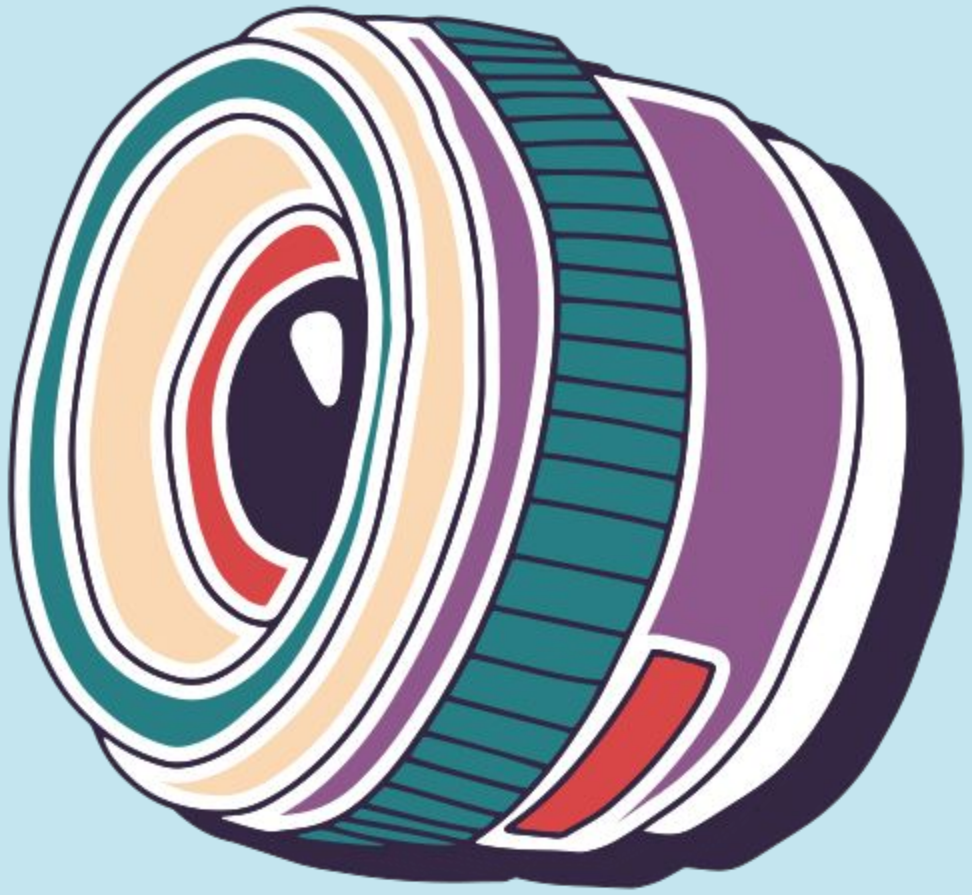
why did we choose this?

- **Economic Accessibility** - Smartphones are far more economically accessible than AR or XR devices.
- **Behavioral Alignment** - Users already take, and store photos, making the transition to automated album sharing seamless.
- **Convenience** - Because smartphones are always present, they enable real-time capture, detection, and sharing without requiring additional setup.
- **Technical Feasibility** - Existing technologies make implementation more realistic.



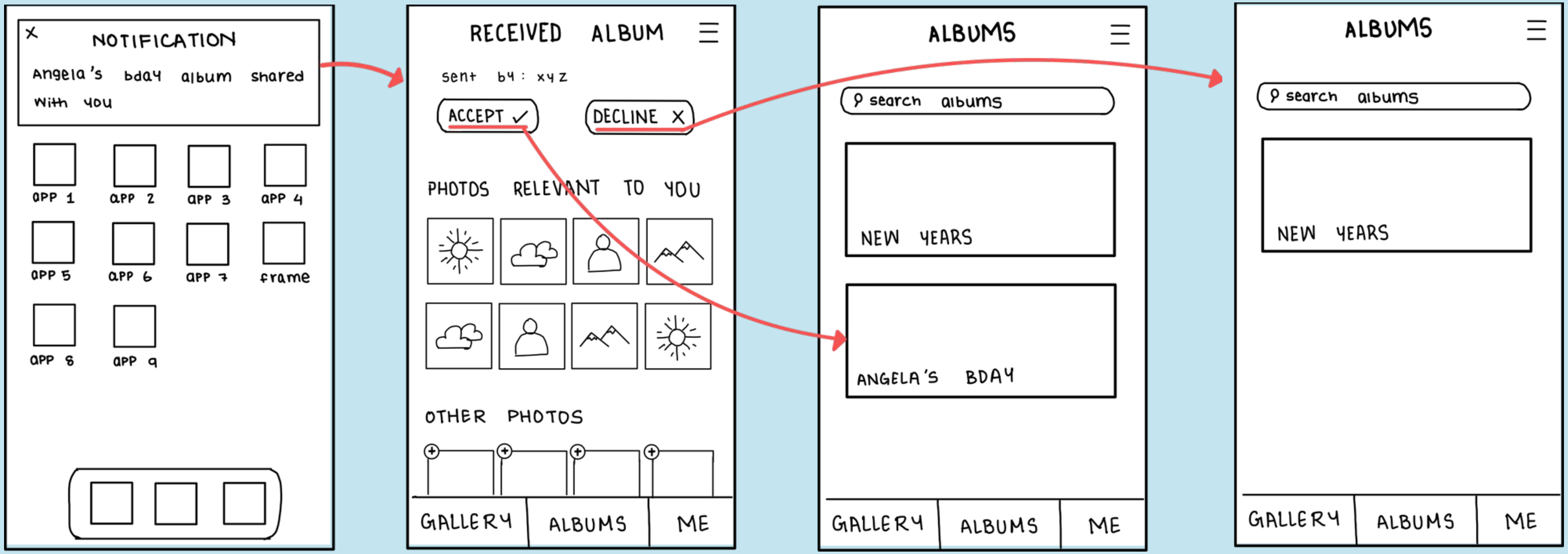
04

task flows



Simple

User receives shared album and accepts or decline it



gets a notification that an album is shared to the user

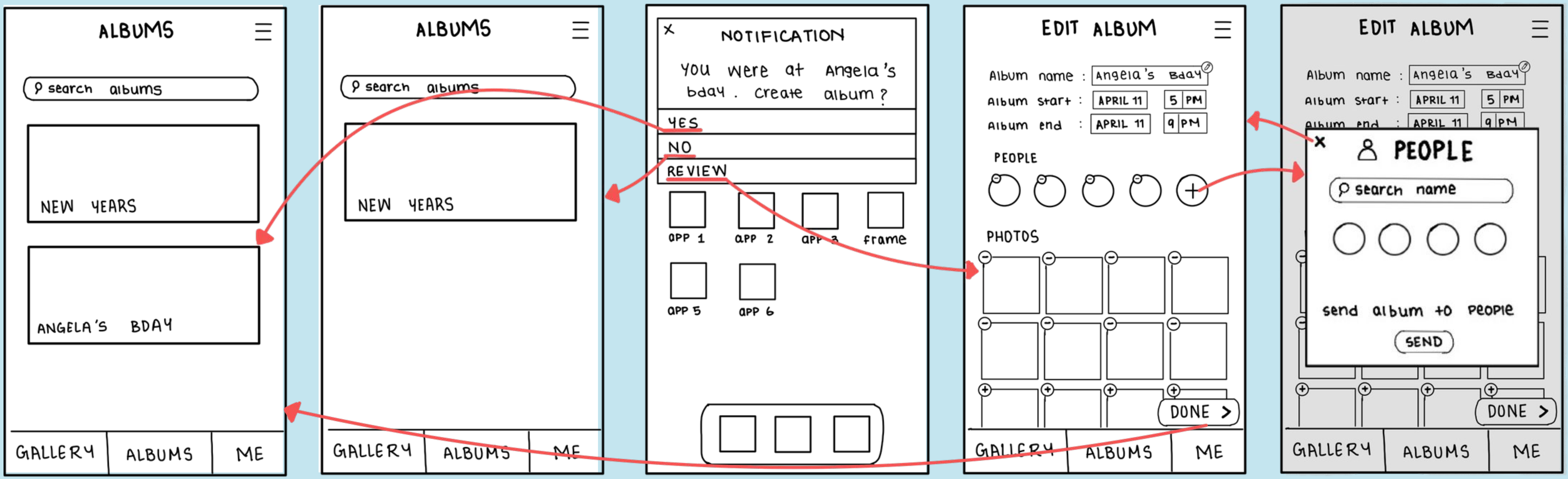
user can view who sent it, all the most relevant photos and other photos of the album and then accept or decline

if accepted, the shared album is in the album screen

if declined, the old albums are in the album screen

moderate

User discovers automatically curated albums by frame



if yes, the new albums is added to the album screen

if no, the old albums are in the album screen

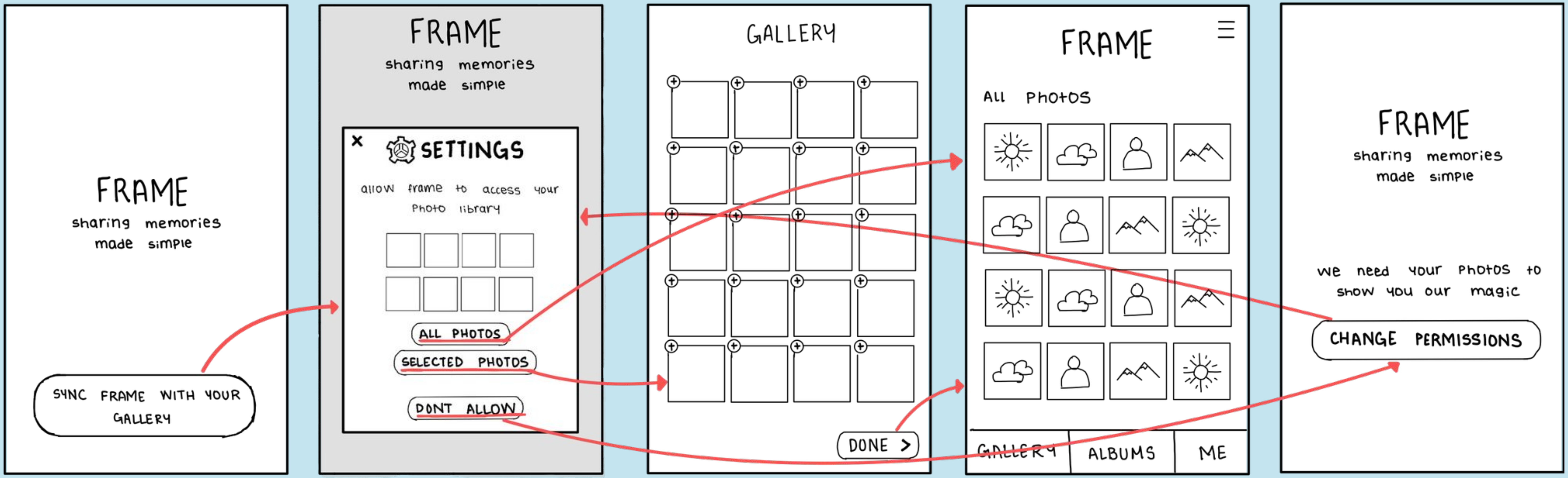
user gets a notification once an album is automatically created, and they have options to choose from for the album

user can review and edit the album by adding or removing photos, changing the timeline and infromation like the name of the album

user can add more people from the suggested or searching them by their name

complex

User imports all or selected photos from his gallery into the app



go to the setting pop up page

user can import all photos or some selected photos

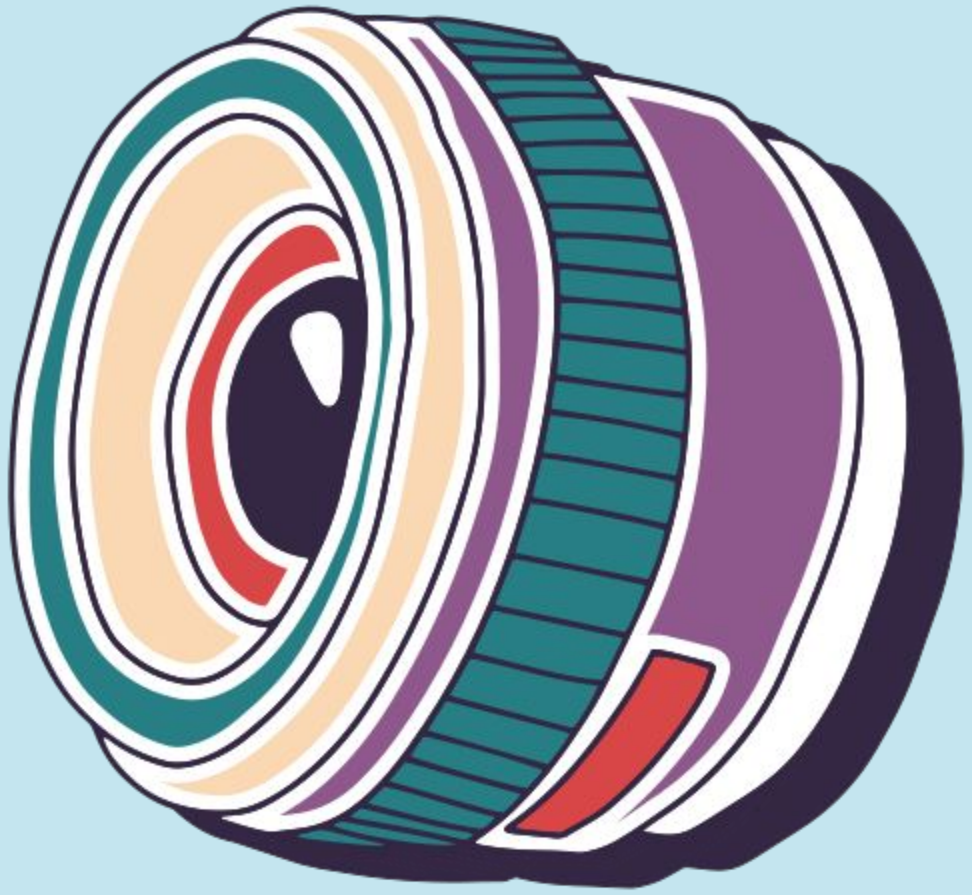
user selects all the photos they want the app to have access to

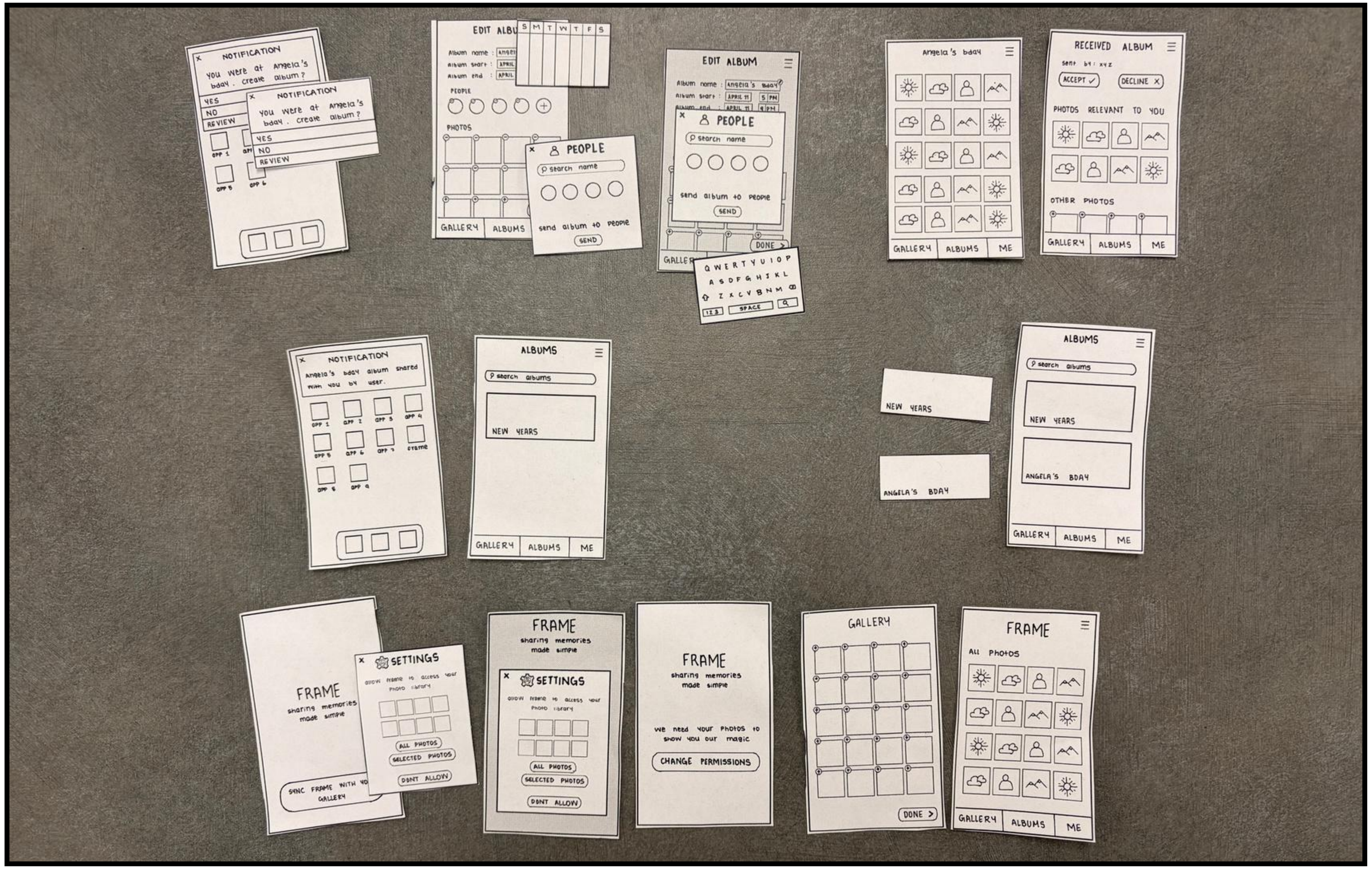
user sees all the selected photos on the dashboard

user can change permission later on or if the user doesnt allow permission

05

Paper Prototype & Testing





cutouts

Construction

Design

Prototype interfaces were first sketched collaboratively on iPads using the GoodNotes app.

Digital sketches were then transferred to paper, and printed/cut out to create consistent low-fidelity screens.

Features

Key flows such as gallery import, auto-album creation, shared album notifications, and album editing were constructed.

Supporting modals were included for realism, including permission prompts, sync screens, participant editing, timeline adjustments, and keyboard inputs.

Function

One team member acted as the “computer,” manually swapping screens and placing interface components in response to user actions.

PARTICIPANTS

Mr. PURPLE

Senior Sociology Major

Extreme User: He is the "historian" of his friend group who takes hundreds of photos at every party.

Mr. RED

International Student, Economics Major

Target User: uses Android because it's standard back home. Has to ask people to email him photos separately.

Mr. GREEN

Gen Z Barista

Average User: Gen Z Barista with 14,000 photos. He takes pictures of everything but never organizes them/

Environment & Apparatus

Test Environment

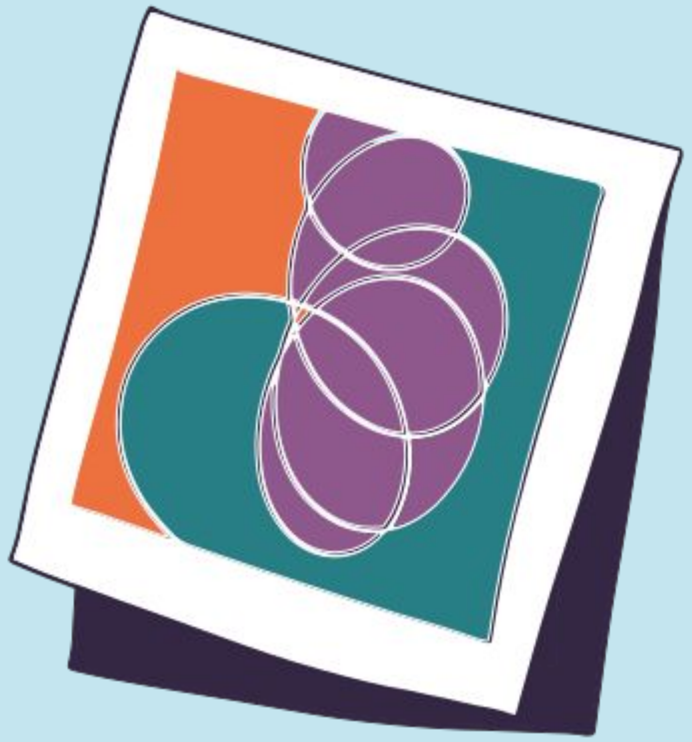
- Conducted in a quiet, neutral setting (e.g., living room and library study room) to simulate a casual environment where users would naturally review photos
- Lighting was controlled to ensure clear visibility of the paper prototype

Apparatus (Equipment):

- Low-Fi Prototype: Physical paper sketches created on index cards/paper to represent mobile screens.
- Recording Device: Smartphone on a stand to record user hands and audio for “Think Aloud” Protocol

Observer Station:

- Laptop for the note-taker to log critical incidents in real-time



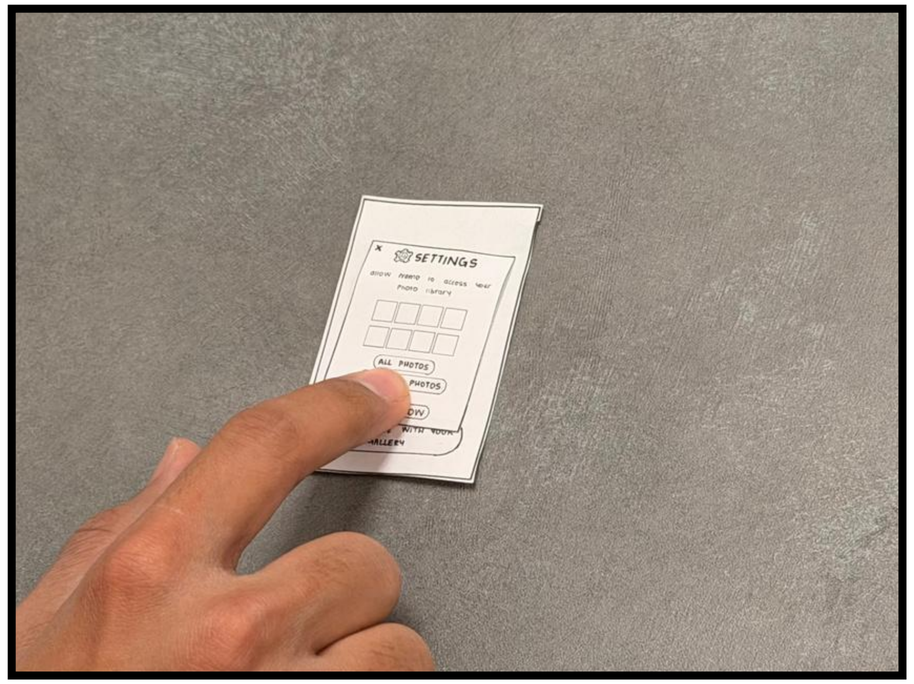
Testing Procedure

Team Member Roles:

- Facilitator (Zeel): Guided the participant, provided task instructions, and prompted them to "think aloud"
- Computer (Gavin): Manually swapped paper screens and simulated system responses (e.g., pop-ups, slide transitions)
- Observer (Rachit): Recorded timing, misclicks, and quotes without interfering.

Session Protocol:

- Onboarding: Signed consent forms and brief explanation of the "Frame" concept.
- Demo: Showed a neutral example (e.g., scrolling) to get them comfortable with paper interaction.
- Task Execution: Users performed the Simple, Moderate, and Complex tasks in order.
- Debrief: Short interview to gather overall impressions and



participants clicking the button, facilitator moving the cutouts

Location: PHELPS 1591

Apparatus: Paper Prototype & Camera

Usability Goals & Measurements

Goal 1: Learnability (Intuitive Navigation)

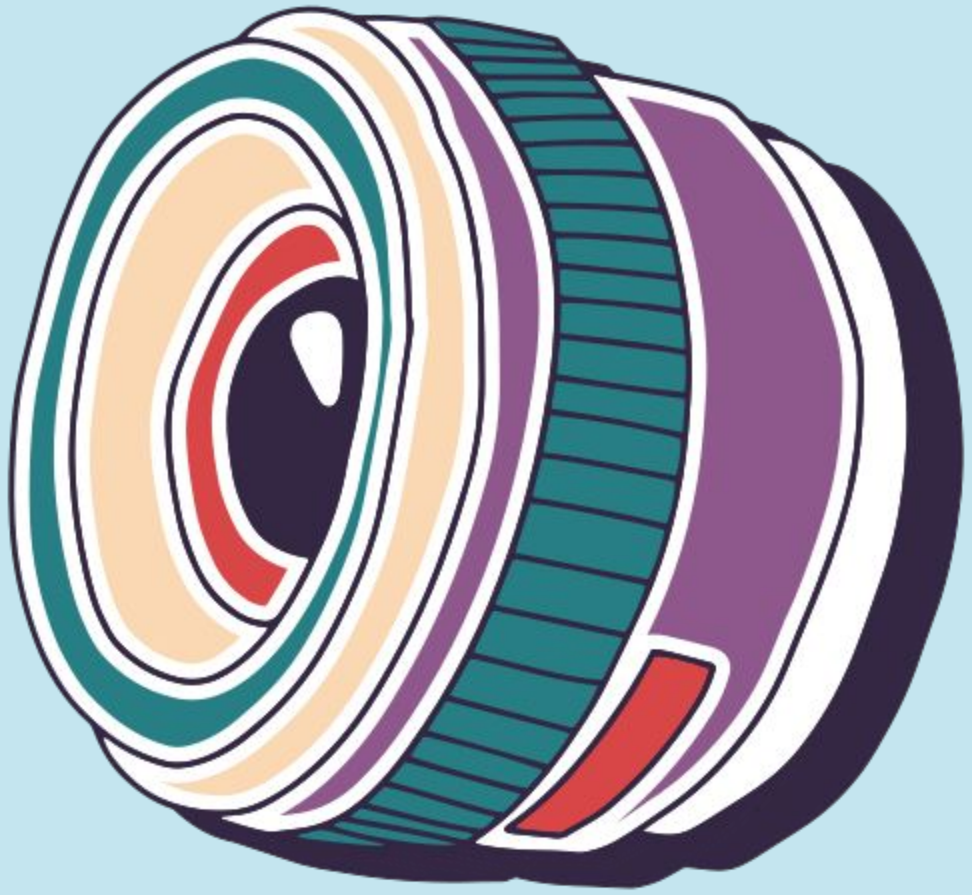
- **Rationale**: Since Frame introduces a new mental model (auto-sharing), users must intuitively understand how to control it.
- Key Measurement (Process Data): Misclick Rate. We counted how many times a user tapped the wrong element (e.g., tapping a face instead of the "Edit" button).

Goal 2: Efficiency (Speed of Sharing)

- **Rationale**: Our problem statement highlights that memory sharing is currently "manual labor". FRAME's core value is "effortless" sharing, so tasks must be faster than current methods.
- Key Measurement (Bottom-Line Data): Time on Task. We

06

Results



Metrics

Participant	Misclicks	Task Completion Time	Mental Effort
Participant 1	1 (Edit Flow)	Fast	2 (Low)
Participant 2	3 (Sync Flow)	Moderate	4
Participant 3	0	Fast	1

Generative Observations

Notification Flow Works Extremely Well

- All users instantly understood the “Angela’s bday album shared” notification
- The “Accept/Decline” simple task had a 100% success rate with zero confusion

High Value Perception of Auto-Creation

- One participant noted, "This saves me so much time picking photos."
- Users appreciated skipping the manual selection process.

Visual Design of “Shared Albums” was Enjoyable

- Users liked seeing the faces (avatars) on the album cover before opening it.

Critical Observations

Permission & Sync Anxiety (Complex Task)

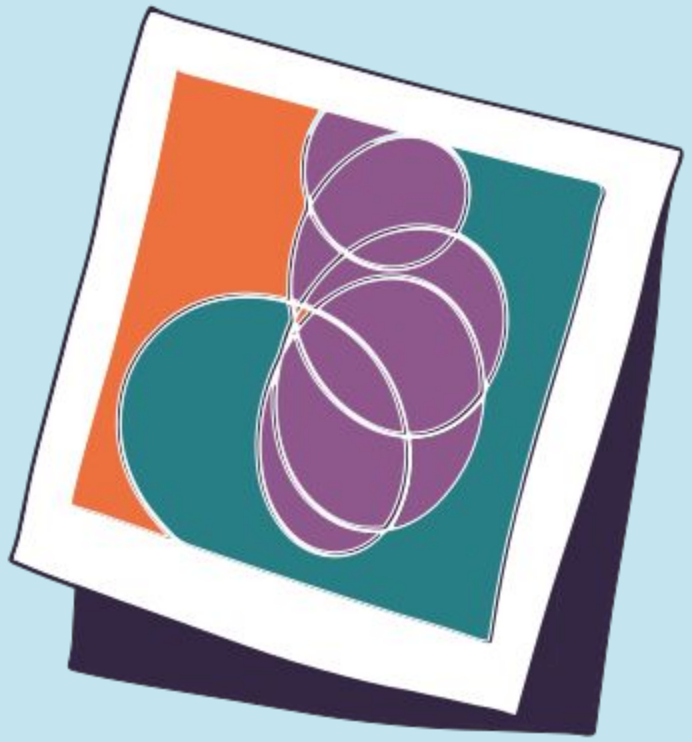
- All three users hesitated at the “Allow Frame to Access Gallery” screen
- They were unsure if “Sync” meant all photos were being uploaded or just the relevant ones
- One user asked, “Is this going to upload my screenshots too?”

“Edit People” Feature Hard to Discover (Moderate Task)

- Users struggled to find the button to remove a person from the auto-generated album
- Two users tried tapping the face icons directly, but the prototype required clicking the “Edit” pencil icon first

Timeline Navigation Difficulties

- When trying to adjust the “Album Start/End Time," users were confused by the slider controls in the paper prototype



Success?

Efficiency

Goal Achieved

- The auto-album creation successfully removed the burden of manual sorting. Time-on-task was very low for the main flows

Trust & Privacy

Goal Not Fully Achieved

- The "Sync" wording caused anxiety. Users need more reassurance about what data is being touched before they feel safe

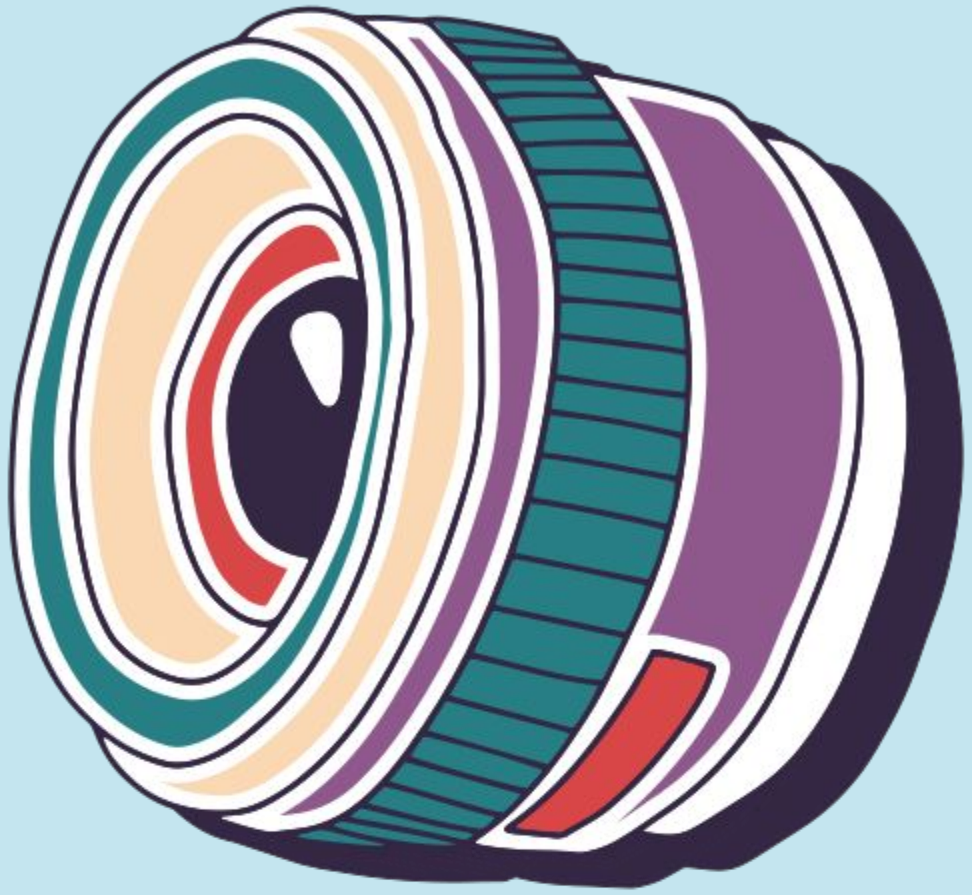
Low Mental Effort

Goal Achieved:

- Average effort was 2.3/5
- Once past the permission screen, the actual sharing and viewing experience was intuitive and “brainless” in a good way

07

Discussion



process data → implication

Users hesitated at the “Allow Frame to Access Gallery” screen because they were unsure what photos would be accessed or uploaded.

We need to design the permission flow to clearly communicate what data is accessed and reassure users about privacy and selective syncing.

process data → implication

Users misunderstood the term “Sync” and assumed the app would upload their entire photo library rather than event-based memories.

We need to replace technical language with clearer, human-centered wording that reflects event-based album creation.

process data → implication

The album notification ("Angela's bday album shared with you") had zero confusion and was immediately understood by all users.

We need to expand proactive notification prompts to drive sharing actions and re-engagement with memories.

process data → implication

Participants perceived auto-album creation as the most useful feature because it removed the burden of manual sorting.

We need to lean further into automation by minimizing manual curation steps and surfacing ready-to-share albums.

Changes

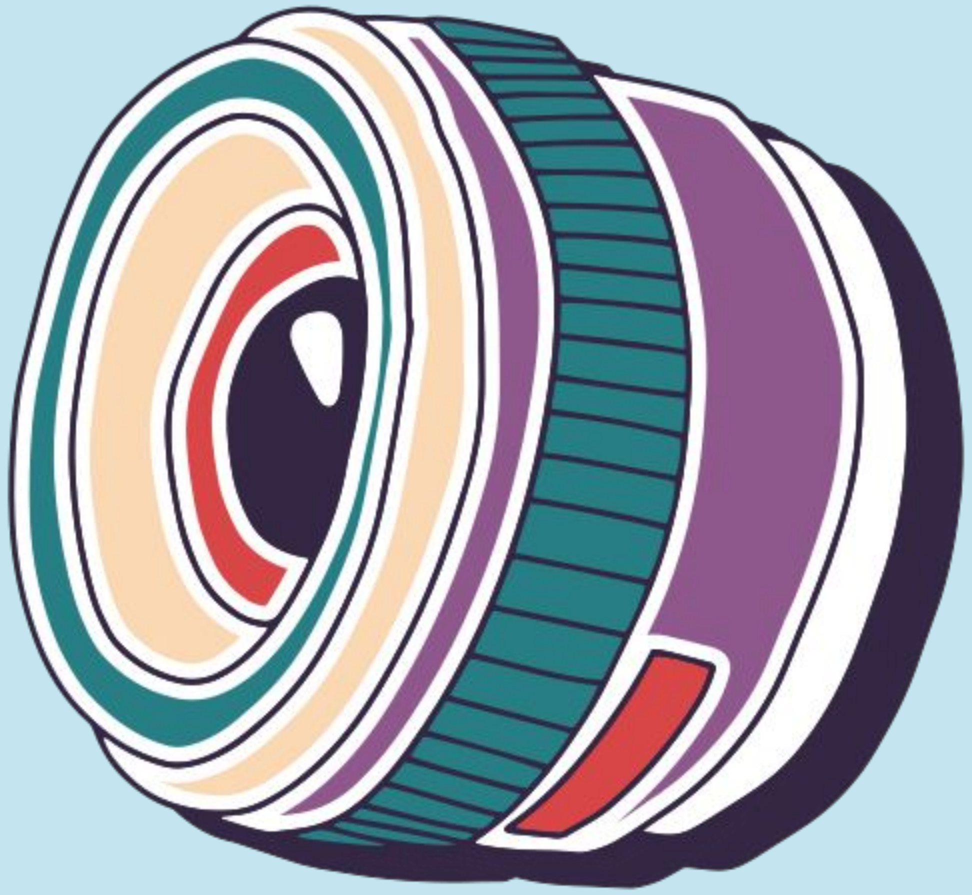
- Clarify gallery permission messaging - add reassuring wordings explaining that only event-related photos are accessed, with an option to modify permissions later.
- Replace the term “Sync” with more human-centered language like “Create Albums from Events” to reduce confusion around full-library uploads.
- Expand notification functionality - include preview thumbnails and action buttons like “Review” or “Share Now” to encourage re-engagement.
- Prioritize auto-generated albums in the dashboard - surface them first and streamline one-tap sharing to reinforce the automation value.

Shortcomings

- Unable to test real facial recognition accuracy - Since this was a paper prototype, we could not evaluate how accurately the system detects participants or clusters event photos.
- Unable to test automatic event detection reliability - We could not simulate real GPS + timestamp clustering to see if albums would be generated at the right moments.
- Unable to test cross-platform album delivery - The prototype could not validate how smoothly albums would open or sync across iOS, Android, and web.
- Unable to test notification timing effectiveness - We could not measure whether album prompts were sent at the most contextually appropriate moments.
- Unable to test real privacy & permission behaviors - Users reacted conceptually to permission screens, but we could not observe long-term trust or opt-out behaviors.
- Unable to test scalability of large photo libraries - The prototype did not simulate performance when processing thousands of photos.

08

Appendix



Phone: Event Auto-Album Creation & Sharing

pros

- Integrates capture -> organization -> sharing in one pipeline - Other solutions fragment the workflow (capture on phone, edit on desktop, share elsewhere), while this approach has all in one.
- Lowest behavioral friction compared to all alternatives - Unlike AR glasses, watches, or desktop, this solution works within an already existing user habit, taking photos on their phone.
- No dependency on synchronous participation - Compared to live collaborative uploads or real-time memory spaces, this concept works even if participants take photos independently and at different times.
- Higher accessibility than hardware-dependent solutions - AR glasses and tablet require specialized devices, while smartphone-based automation is economically and technologically accessible.
- Eliminates manual labor present in other phone and desktop concepts - Solutions like drag-and-drop distribution workspaces or collaborative editing require intentional effort, whereas auto-album creation shifts to the system.
- Greater scalability for large social events - Watch proximity sharing or tablet co-curation work best in small group settings, while automated clustering supports large gatherings.
- Works post-event rather than requiring in-event engagement - Live upload and real-time shared memory spaces demand attention during the event, whereas this concept lets the user be present in the moment.

Phone: Event Auto-Album Creation & Sharing

cons

- Less intentional than manual curation tools - Desktop editing workspaces and drag-and-drop distribution allow precise storytelling, whereas automation may feel less personally crafted.

- Lower experiential novelty compared to AR solutions - Location-based viewing and immersive spatial galleries offer memorable, futuristic interactions that this phone solution does not.

- Reduced real-time engagement - Live collaborative upload and shared AR memory spaces foster in-moment connection, while this concept operates retrospectively.

- Heavier reliance on AI inference - Unlike proximity triggers or manual sorting, this system depends on facial recognition and clustering accuracy to function well.

- Privacy concerns amplified compared to opt-in systems - Temporary sharing or selective sharing tools rely on explicit user choice, whereas automated detection introduces passive data processing.

- Less playful or socially performative - Tablet curation tables or collaborative editing sessions create shared social activities, while automation is largely invisible.

- Potential over-automation - Users who enjoy organizing memories themselves may prefer desktop or manual

AR Glasses: Live Group Memory Space

pros

- Real-time collective presence vs post-event reflection - Unlike auto-album phone solutions that act after memories are captured, this concept allows users to witness memories forming live, creating a sense of shared authorship.
- Strongest social co-experience layer - Compared to desktop collaborative editing (remote) or tablet curation (after the event), AR enables collaboration duringthe event itself.
- Spatial anchoring enhances recall - Phone galleries and desktop folders rely on chronological sorting, whereas spatial placement ties memories to environmental cues, improving long-term recall.
- Contextual Immersion: Viewing memories at the exact location they occurred creates a stronger emotional connection than scrolling through a disconnected gallery on a phone.
- Hands-Free Experience: Solves the "app switching" pain point by allowing users to interact with memories without breaking engagement with the real world (e.g., while hiking).
- Spatial Mental Model: Leveraging spatial memory reduces the cognitive load of searching through folders.
- Novelty & Differentiation: Fulfills the assignment goal of "novel realizations," distinguishing Frame from standard competitors like Google Photos or iCloud.
- Eliminates screen mediation entirely - All phone and tablet solutions still require users to look down at screens, breaking immersion in the event.

AR Glasses: Live Group Memory Space

cons

- High Barrier to Entry: The target user base (students/young adults) generally does not own AR glasses, which severely limits adoption compared to a smartphone app.
- Location Dependency: The core value fails if users cannot physically revisit a location (e.g., a one-time vacation spot). A phone app works anywhere.
- Hardware dependency vs universal phone access - All phone solutions work instantly for billions of users, whereas AR glasses adoption is still niche and cost-prohibitive.
- Battery + compute limitations - Continuous spatial rendering and multi-user syncing is far heavier than background album clustering on phones.
- Environmental constraints - Desktop and phone tools work indoors, outdoors, anytime — AR spatial overlays require safe, trackable physical environments.
- Social Friction: Gesturing at invisible objects in public can cause social awkwardness and raise privacy concerns for bystanders who don't know if they are being recorded.
- Lack of Tactile Feedback: "Air tapping" is less precise than a phone screen, potentially making simple tasks (like selecting 1 specific photo) more frustrating and error-prone.

Aggregated Critical Incidents

Explanation	Task	Description Of Event	Type	Severity
"Is this going to upload my screenshots too?" Users paused before proceeding due to privacy concerns.	Sync / Permission Setup	Multiple users hesitated at the gallery permission screen and were unsure what photos the app would access or upload.	Negativ e	4
User initially tapped face avatars directly instead of using the edit control, causing navigation friction	Auto-Album Review / Edit Flow	One user misclicked while trying to edit participants in the auto-generated album.	Negativ e	2

Aggregated Critical Incidents

Explanation	Task	Description Of Event	Type	Severity
“This saves me so much time picking photos.” Participants valued skipping manual sorting.	Auto-Album Creation Experience	Users expressed strong appreciation for automatic album generation after events.	Positive	0
Participants were unsure how timeline edits affected album content.	Timeline Adjustment / Album Range	Users were confused when adjusting album start/end time using slider controls.	Negative	2
Fast task completion times + low reported mental effort	Overall Sharing Flow Completion	Most users completed the main sharing flow quickly with	Positive	0

Script

This is a prototype for our app, Frame, that is meant to make sharing memories easier. The general idea is that Frame automatically creates shared photo albums based on events you attend, and helps you review, edit, and share those memories without having to manually collect photos from friends.

Gavin will be the computer, so they are just making these papers work like the technology would, and Rachit is our observer. Neither of them will speak so you can just pretend they don't exist.

I will be asking you for your thoughts as you walk through the app, but I won't help you navigate anything so that we can see your natural instincts.

Demo - **Show how one would "click" buttons and how that would navigate to a new page/change the state of the page. You will be completing three tasks.**

Let's start with the first task: "Accept and view a shared album."
You will receive a notification that an album from an event you attended has been shared with you. Please open the notification and explore the album.
Do task

Now, the second task is "Review and edit an auto-created album."
Please open the album that Frame has automatically created for you, review the photos, and edit the people included in the shared album.

continued ->

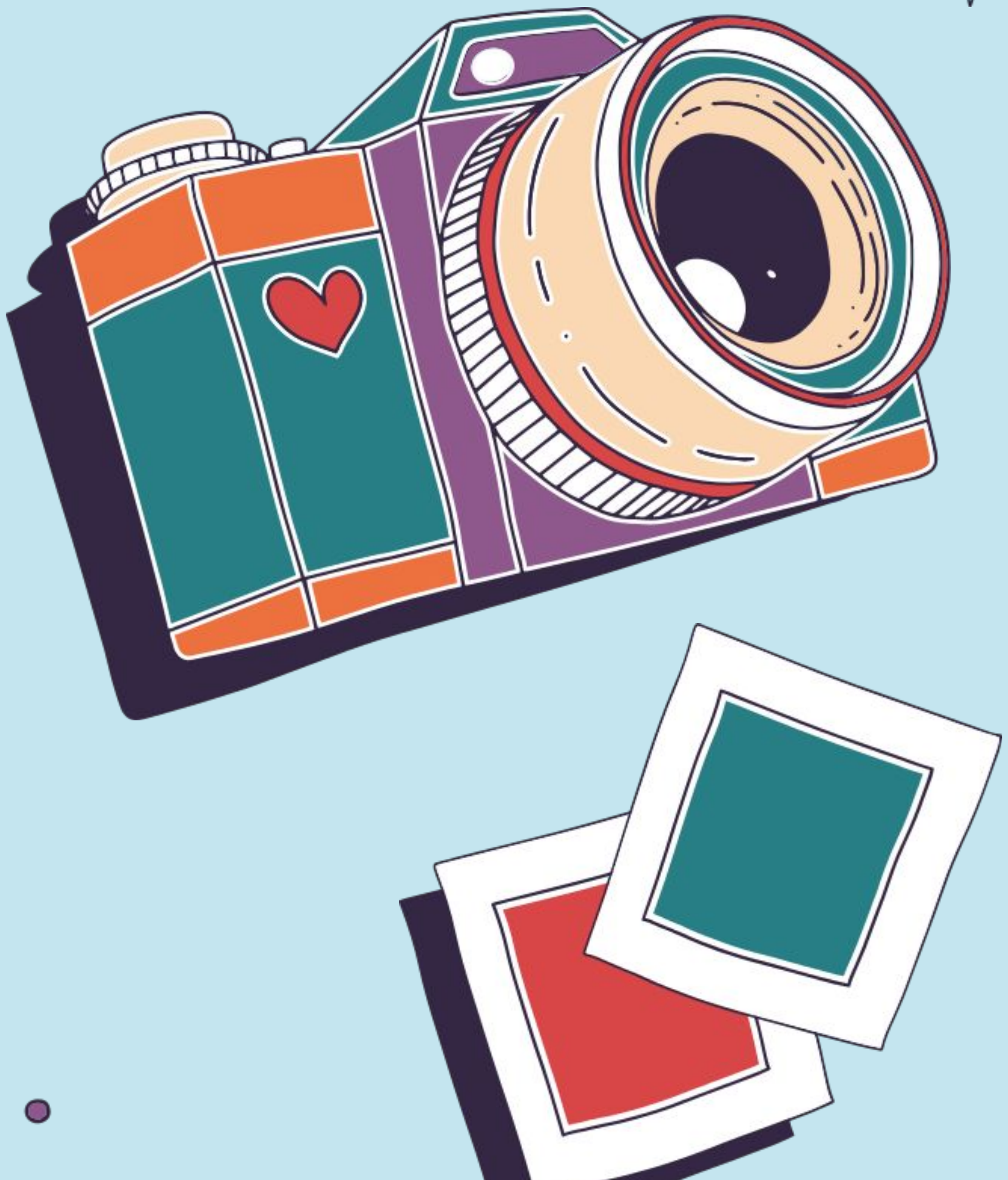
Script

Finally, the third task is “Set up photo sync permissions.”
Please go through the permission flow and decide what level of gallery access you would like to give the app.
Do task

If they messed up on any tasks:
I am now going to ask you to do the xyz task one more time. *Repeat the messed up tasks.*

Ask: What would you say your mental effort was during this process? More specifically, on a scale of 1–5 how much mental energy/focus did you need to exert?

Thanks for participating in our testing. Do you have any general feedback, thoughts, or suggestions?



thank you!